

Predicting Adverse Pregnancy Outcomes Using Machine Learning: An Analysis of the nuMoM2b Dataset

by

Hetanshee Shah

Advisor:

Dr. Mandy Korpusik
Faculty of the Department of
Computer Science
Loyola Marymount University

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Abstract

This thesis aims to identify key predictors of adverse pregnancy outcomes, maternal mental health, and breastfeeding outcomes using machine learning models applied to the Nulliparous Pregnancy Outcomes Study: Monitoring Mothers-to-Be (nuMoM2b) dataset. From the initial nuMoM2b cohort of 10,038 nulliparous women, a final analytical sample of 5,720 participants was derived after applying exclusion criteria. Predictive models were developed using features from multiple domains, including demographics, maternal health, sleep, diet, and physical activity. Exploratory data analysis techniques such as histograms, heatmaps, clustering, and box plots were used for feature selection. Data preprocessing steps included one-hot encoding for categorical variables, scaling for numerical variables, and SMOTE to handle class imbalance. Logistic regression and other machine learning models were trained on individual domains and then combined to predict mental health risks, baby health outcomes, and breastfeeding. Sleep features emerged as strong predictors, with the combined model incorporating sleep and other domains achieving an accuracy of 85% in predicting maternal health outcomes, albeit with a smaller sample size ($n=218$). Models for breastfeeding outcomes showed limited predictive power, suggesting the need for additional features and further research. Feature importance analysis identified key variables from each domain contributing to the predictive power of the models. Machine learning models applied to the nuMoM2b dataset provide valuable insights into factors influencing adverse pregnancy outcomes and maternal health. Sleep patterns, in combination with demographic, health, and behavioral features, significantly improve the ability to predict mental health risks post-pregnancy. The identification of key predictors can guide targeted interventions and risk assessment strategies to improve maternal and child health outcomes. However, the limited success in predicting breastfeeding highlights the complexity of this outcome and the need for further research.

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Contents

1	Introduction	10
1.1	Contributions & Outline	10
2	Literature Review	12
2.1	Maternal Mental Health and Pregnancy Outcomes	12
2.1.1	Impact of Anxiety and Depression on APOs	12
2.1.2	Sociodemographic Influences	13
2.2	Physical Health Predictors in Pregnancy	14
2.2.1	Sleep Health and Gestational Outcomes	14
2.2.2	Cardiovascular Metrics in Pregnancy	14
2.3	Machine Learning in Predicting Health Outcomes	15
2.3.1	Overview of Machine Learning Models	15
2.3.2	Applications in Maternal and Neonatal Health	17
2.4	Gaps in Current Research	18
2.5	Summary	20
3	Methodology	21
3.1	Research Design and Data Source	21
3.2	Data Source: The nuMoM2b Dataset	23
3.2.1	Participant Selection Criteria	23
3.2.2	Data Collection Procedures	24
3.3	Variables and Measures	25
3.3.1	Dependent Variables (Adverse Pregnancy Outcomes)	25
3.3.2	Independent Variables	25
3.4	Data Preprocessing	27
3.4.1	Handling Missing Data	27
3.4.2	Variable Encoding and Scaling	27

3.4.3	Sample Size Considerations and Data Partitioning	28
3.5	Exploratory Data Analysis (EDA)	28
3.6	Individual Domain-Specific Models	31
3.6.1	Paternal Demographics Model	33
3.6.2	Maternal Demographics Model	33
3.6.3	Maternal Health Model	34
3.6.4	Food and Nutrition Model	34
3.6.5	Health Knowledge Model	34
3.6.6	Physical Activity Model	34
3.6.7	Sleep Model	35
3.7	Model Evaluation Metrics	35
3.7.1	Classification Performance Metrics	35
3.7.2	ROC Analysis and AUC Evaluation	36
4	Results	38
4.1	Descriptive Statistics	38
4.1.1	Participant Demographics	38
4.1.2	Distribution of Key Variables	39
4.2	Model Performance	41
4.2.1	Predicting Maternal Mental Health Outcomes	41
4.2.2	Predicting Neonatal Health Outcomes	42
4.3	Summary of Key Findings	44
5	Discussion	46
5.1	Interpretation of Results	46
5.1.1	Maternal Mental Health Predictions	46
5.1.2	Neonatal Health Prediction Challenges	47
5.2	Comparison with Existing Literature	48

5.3	Implications for Clinical Practice and Public Health	50
5.3.1	Sample Size Constraints	51
5.3.2	Data Limitations	51
5.4	Future Directions	52
6	Conclusion	54

List of Figures

- 1 Flow diagram of study cohort derivation. The initial nuMoM2b cohort (n=10,038 participants) was refined to those with complete sociodemographic, physical health, and pregnancy outcome data (n=7,790). Of these, 5,720 participants completed all mental health assessments. The smallest subset (n=218) represents participants who additionally completed overnight sleep studies with valid actigraphy measurements. 22
- 2 Histograms of numerical features illustrating the distribution of key variables, including dietary intake (e.g., total calories, protein, and carbohydrate consumption), sleep metrics, stress levels, and other health-related metrics. The histograms highlight variability across the dataset, providing insights for data preprocessing and normalization. 29
- 3 Correlation matrix of numerical features visualizing the relationships between various predictors, such as dietary intake, sleep health, and stress metrics. The color gradient indicates the strength and direction of correlations, with red representing positive correlations and blue representing negative correlations. This matrix aids in identifying multicollinearity and selecting independent predictors for machine learning models. 30
- 4 Flowchart illustrating the process of data preprocessing and analysis. The diagram details each step from raw data ingestion to final model evaluation. 32
- 5 Comparison of ROC curves for four models: Logistic Regression, Random Forest, XGBoost, and Stacking Classifier. The Random Forest model achieved the highest AUC (0.86), closely followed by the Stacking Classifier (AUC = 0.85) and XGBoost (AUC = 0.81). Logistic Regression had the lowest AUC (0.75). The dashed diagonal line represents a random classifier (AUC = 0.5). 37

6	Feature importances derived from the Random Forest model. The <i>Has-</i> <i>sle/Uplift intensity ratio</i> and sleep fragmentation metrics are the most sig- nificant predictors, followed by psychological resilience and specific dietary factors. The remaining variables exhibit progressively lower importance in predicting the outcome.	43
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List of Tables

1	Summary Statistics for Mother's Age	38
2	Frequency Distribution of Race/Ethnicity	39
3	Income Level Distribution and Mental Health Outcomes	39
4	Model Performance Metrics for Mental Health Prediction	42
5	Model Performance Metrics for Breastfeeding Intention Prediction	42
6	Description of key variables used in the analysis	61
6	Description of key variables used in the analysis	64
7	Sleep Parameter Analysis Between Normal and Adverse Outcome Groups . .	65
8	Psychological Parameter Analysis Between Outcome Groups	65

1 Introduction

Pregnancy is a critical period in a woman’s life that involves significant physiological, psychological, and social changes. While most pregnancies progress without complications, a substantial proportion of women experience adverse pregnancy outcomes (APOs) that can have profound and lasting effects on both maternal and child health. APOs, such as preterm birth, gestational diabetes, preeclampsia, and low birth weight, are associated with increased morbidity and mortality, as well as long-term health consequences for mothers and their offspring [36, 21].

Despite advances in prenatal care and obstetric practice, the prevalence of APOs remains a significant public health concern. In the United States, preterm birth affects approximately 10% of all deliveries, while gestational diabetes and preeclampsia complicate around 6-8% and 3-5% of pregnancies, respectively [14]. These outcomes not only pose immediate risks to the mother and newborn but also contribute to health disparities and increased healthcare costs [29].

Identifying risk factors and developing predictive models for APOs has been a major focus of maternal health research. Traditional approaches have emphasized clinical and demographic factors, such as maternal age, parity, pre-existing medical conditions, and socioeconomic status [19]. However, growing evidence suggests that a more comprehensive, multidimensional approach incorporating psychological, nutritional, and sleep-related factors may improve our ability to predict and prevent APOs [36, 24].

1.1 Contributions & Outline

The thesis aims to address gaps in current knowledge by applying a comprehensive, multidimensional approach to predicting APOs using data from the Nulliparous Pregnancy Outcomes Study: Monitoring Mothers-to-Be (nuMoM2b) cohort [14]. We investigate the associations between maternal psychological factors (stress, anxiety, depression), nutritional

factors (dietary patterns, nutrient intakes), sleep parameters (duration, quality, timing), and APOs (preterm birth, gestational diabetes, preeclampsia, low birth weight).

Furthermore, we develop and validate machine learning models that integrate these multi-dimensional predictors to identify women at high risk of APOs. By evaluating the relative importance of psychological, nutritional, and sleep-related factors in shaping pregnancy outcomes, we explore potential interactions between these domains.

This thesis aims to provide insights into the mechanisms linking maternal health factors to APOs and inform the development of targeted interventions to promote healthy pregnancies and optimal outcomes for mothers and infants.

The remainder of this thesis is organized as follows: Chapter 2 provides a comprehensive review of the existing literature on maternal health factors and APOs, highlighting key findings and gaps in current knowledge. Chapter 3 describes the methods used in the thesis, including the nuMoM2b dataset, variable definitions, and statistical analyses. Chapter 4 presents the results of our analyses, including descriptive statistics, machine learning model performance, and feature importance rankings. Chapter 5 discusses the implications of our findings for clinical practice and public health, and outlines directions for future research. Finally, Chapter 6 summarizes the main conclusions of the thesis and highlights its contributions to the field of maternal and child health.

2 Literature Review

The prediction of adverse pregnancy outcomes (APOs) is a critical area of maternal and neonatal health research, as these outcomes can have long-lasting impacts on the well-being of both mother and child. Current literature highlights a diverse array of predictors—from mental health and sociodemographic factors to physical health and sleep patterns—underscoring the need for multidimensional approaches to risk assessment. Despite significant advancements, several gaps persist in understanding these predictors’ interactions and leveraging modern analytical tools, such as machine learning, to enhance prediction accuracy. This review synthesizes the existing body of knowledge, identifying key predictors and methodologies while emphasizing their implications for this thesis.

2.1 Maternal Mental Health and Pregnancy Outcomes

Maternal mental health is increasingly recognized as a critical determinant of pregnancy outcomes. Anxiety and depression during pregnancy are linked to adverse pregnancy outcomes (APOs) such as preterm birth, low birth weight, and developmental challenges in infants. Studies underscore the biological and psychological pathways through which these mental health conditions affect pregnancy, highlighting the need for early detection and intervention to mitigate risks [34, 5].

2.1.1 Impact of Anxiety and Depression on APOs

Anxiety and depression have profound implications for both maternal and neonatal health. High levels of anxiety, as measured by instruments like the State-Trait Anxiety Inventory (STAI), have been associated with increased risks of preterm birth and low birth weight [34]. Similarly, depressive symptoms during pregnancy, assessed using the Edinburgh Postnatal Depression Scale (EPDS), have been correlated with impaired fetal growth and developmental delays [5]. These findings are supported by longitudinal data from large cohort studies,

which suggest that untreated maternal mental health conditions can exacerbate stress-related physiological responses, such as elevated cortisol levels, contributing to adverse outcomes [14].

Emerging research emphasizes the interplay between mental health and other physiological factors. For example, stress and depressive symptoms often coincide with poor sleep quality and disrupted circadian rhythms, which further amplify risks for APOs [38]. These multifaceted interactions highlight the importance of integrating mental health variables with other predictors in comprehensive risk assessments.

2.1.2 Sociodemographic Influences

Sociodemographic factors often mediate the impact of maternal mental health on pregnancy outcomes. Maternal age, race, ethnicity, income, education level, and access to healthcare significantly shape health trajectories [17]. For instance, lower socioeconomic status and limited healthcare access are associated with higher levels of stress and anxiety during pregnancy, which, in turn, increase the likelihood of APOs [25]. Furthermore, disparities in mental health outcomes and healthcare access among racial and ethnic groups underscore the need for culturally sensitive interventions and equitable healthcare policies.

These findings suggest that addressing maternal mental health in isolation may not suffice; instead, a holistic approach that considers sociodemographic contexts is necessary. This aligns with recommendations from the nuMoM2b study, which advocates for the inclusion of diverse populations in research to better understand the nuanced effects of mental health and sociodemographic factors on pregnancy outcomes [14].

In summary, maternal mental health is a pivotal factor in predicting APOs, with anxiety, depression, and stress emerging as key variables. However, the intricate interplay between mental health and sociodemographic influences necessitates multidimensional approaches for effective risk prediction and intervention. By integrating these insights, our research aims to enhance predictive accuracy and inform targeted clinical strategies.

2.2 Physical Health Predictors in Pregnancy

Physical health predictors play a pivotal role in determining pregnancy outcomes. Variables such as sleep health, cardiovascular metrics, and pre-existing conditions influence the likelihood of adverse pregnancy outcomes (APOs). Recent research has deepened our understanding of how these factors operate independently and interact with mental health and sociodemographic variables to shape maternal and neonatal health trajectories.

2.2.1 Sleep Health and Gestational Outcomes

Sleep health has emerged as a critical but often underexplored predictor of APOs. Poor sleep quality, irregular sleep timing, and inadequate sleep duration are linked to an increased risk of preterm birth, gestational diabetes mellitus (GDM), and abnormal gestational weight gain (GWG) [11, 38]. For instance, [15] demonstrated that multidimensional sleep health, including domains such as efficiency, duration, and regularity, is a stronger predictor of GWG than isolated sleep parameters. Actigraphy-based studies reinforce these findings, highlighting the relationship between poor sleep continuity and elevated risks for both maternal and neonatal complications [11].

Sleep-disordered breathing (SDB) further exacerbates risks, with studies indicating associations between SDB and low birth weight and other APOs [31]. These findings underscore the necessity of incorporating objective sleep measures, such as actigraphy and polysomnography, into maternal health assessments. The thesis builds on this foundation by integrating sleep health variables into predictive models, improving adverse outcome predictions’ accuracy in smaller but well-characterized datasets.

2.2.2 Cardiovascular Metrics in Pregnancy

Cardiovascular health during pregnancy offers another lens for understanding APOs. Hypertensive disorders, such as preeclampsia and gestational hypertension, are major contributors to maternal morbidity and mortality [26]. Research from the nuMoM2b Heart Health Study

reveals that APOs, including preeclampsia and gestational hypertension, not only impact immediate pregnancy outcomes but also have long-term implications for cardiovascular health in mothers [14]. For example, women with hypertensive disorders during pregnancy face a higher risk of developing chronic hypertension and cardiovascular diseases later in life [13].

Moreover, the association between cardiovascular metrics and APOs extends to neonatal outcomes. Studies have shown that maternal hypertension can lead to low birth weight, preterm birth, and increased neonatal intensive care admissions [13]. These insights emphasize the importance of monitoring cardiovascular health throughout pregnancy as part of a comprehensive strategy for mitigating risks.

The integration of physical health predictors, such as sleep and cardiovascular metrics, with mental health and sociodemographic factors, provides a more holistic understanding of APO risks. This multidimensional approach aligns with the findings of recent studies, which advocate for comprehensive health assessments to improve prediction models and early intervention strategies [15]. By leveraging these insights, our research seeks to enhance the predictive power of machine learning models, offering actionable guidance for clinicians and policymakers in optimizing maternal and neonatal outcomes.

2.3 Machine Learning in Predicting Health Outcomes

Machine learning (ML) has emerged as a transformative tool in healthcare, offering sophisticated methods for analyzing complex datasets and improving predictive accuracy for various health outcomes. In maternal and neonatal health, ML models enable the integration of diverse predictors—ranging from clinical measurements to behavioral and sociodemographic factors—to enhance the prediction of adverse pregnancy outcomes (APOs).

2.3.1 Overview of Machine Learning Models

A significant challenge in healthcare prediction tasks is the inherent imbalance in outcome distributions, where adverse events are typically less common than normal outcomes. To

address this challenge in our dataset, we employed the Synthetic Minority Over-sampling Technique (SMOTE), which creates synthetic examples of the minority class by interpolating between existing minority class instances. This approach helped balance our dataset and improve model performance on the less common but critically important adverse outcomes.

Building on this balanced dataset, we employed several supervised machine learning algorithms to predict maternal health outcomes. Logistic regression served as our baseline model due to its effectiveness when the relationship between predictors and the outcome is linear [16]. To capture more complex patterns in the data, we incorporated tree-based ensemble methods, namely random forests and gradient boosting machines (GBMs). Random forests, which construct multiple decision trees and combine their outputs, are known for their ability to improve predictive accuracy and control overfitting [3]. XGBoost, a highly optimized GBM implementation, was selected for its speed, performance, and effectiveness in handling missing data and large feature sets [4].

For capturing potential nonlinear relationships and interactions among predictors, we extended our analysis to include support vector machines (SVMs). These models are particularly powerful for classification tasks with high-dimensional data, as they seek to find the hyperplane that best separates data points of different classes [6]. To further enhance our predictive capabilities, we implemented a stacking ensemble approach, which combines the predictions of multiple base models (logistic regression, random forest, and XGBoost) using a meta-classifier. This strategy aimed to leverage the strengths of each model and improve overall predictive performance [39].

The selection and implementation of these machine learning models was guided by the need to balance predictive accuracy with interpretability. While complex models like XGBoost and stacking ensembles were expected to offer higher performance, the inclusion of logistic regression and random forests ensured that the decision-making process remained interpretable to stakeholders in clinical settings [30].

Through this comprehensive approach, we sought to thoroughly evaluate the predictive

utility of our features and identify the most robust method for predicting pregnancy health outcomes.

2.3.2 Applications in Maternal and Neonatal Health

Machine learning models have been increasingly applied to predict APOs, leveraging large datasets to identify patterns and risk factors that may not be apparent through traditional statistical methods. Studies have utilized ML to predict outcomes such as preterm birth, gestational diabetes mellitus (GDM), preeclampsia, and low birth weight.

For instance, [8] employed machine learning algorithms to predict preterm birth using electronic health records, achieving higher predictive accuracy compared to traditional methods. Their models incorporated a wide range of features, including demographic information, medical history, and laboratory results, demonstrating the potential of ML in synthesizing complex data for APO prediction.

Similarly, [22] used ML models to predict GDM by integrating clinical variables and genetic information. Their gradient-boosting model outperformed traditional risk assessment methods, highlighting the value of ML in combining diverse data types.

In the context of sleep health and its impact on pregnancy outcomes, [31] explored the relationship between sleep-disordered breathing and gestational diabetes using logistic regression models. While not explicitly an ML approach, their study provides a foundation for incorporating sleep variables into predictive models. Building on this, ML algorithms can effectively handle the non-linear relationships and interactions between sleep health, mental health, and other physiological factors.

Moreover, addressing health disparities through ML has been a focus of recent research. [28] discussed how bias can be introduced into algorithms if not properly addressed, particularly when models are trained on datasets that are not representative of diverse populations. This is especially relevant in maternal health, where racial and socioeconomic disparities significantly affect outcomes. Ensuring that ML models are developed and validated on diverse

datasets is crucial for equitable healthcare delivery.

This thesis leverages these advancements by applying ML models, specifically gradient-boosting machines like XGBoost, to predict APOs using a comprehensive set of predictors, including mental health assessments, sleep health metrics, cardiovascular data, and sociodemographic factors. By integrating these variables, we aim to capture the complex interplay between different domains contributing to APOs.

We also address model interpretability challenges by employing techniques such as SHAP (SHapley Additive exPlanations) values, which help explain the contribution of each feature to the model’s predictions [23]. This approach enhances the clinical applicability of our models by providing insights that can inform targeted interventions.

In summary, machine learning offers powerful tools for improving the prediction of adverse pregnancy outcomes. By integrating diverse datasets and addressing issues of interpretability and bias, ML models can contribute significantly to maternal and neonatal health, guiding early interventions and personalized care strategies.

2.4 Gaps in Current Research

Despite the significant progress made in understanding adverse pregnancy outcomes (APOs) and their predictors, several gaps remain in the existing body of research. One critical limitation is the lack of integrated approaches that encompass the multifaceted dimensions of maternal health, including mental health, sleep patterns, cardiovascular metrics, and sociodemographic factors. While studies have independently explored these domains, such as [15] work on multidimensional sleep health and gestational weight gain, or [31]’s research on sleep-disordered breathing, there is limited work synthesizing these variables into a unified predictive framework.

Another prominent gap lies in the underrepresentation of diverse populations in many datasets. Studies like [28] have highlighted the risks of bias in machine learning (ML) models trained on homogenous datasets, which can result in inequitable healthcare delivery. This is

particularly relevant in maternal health, where racial, ethnic, and socioeconomic disparities significantly impact pregnancy outcomes. Despite these known disparities, many studies fail to account for or adequately address these variables, limiting the generalizability and equity of their findings.

Additionally, while ML models are increasingly applied to predict APOs, their interpretability and clinical applicability remain challenges. Complex models, such as gradient boosting machines and neural networks, often function as “black boxes,” offering limited insight into how predictions are made. Although techniques like SHAP values [23] provide interpretability, their adoption in clinical practice is still evolving. This underscores the need for models that balance predictive performance with interpretability, ensuring that healthcare providers can trust and utilize these tools effectively.

The integration of sleep metrics into APO prediction is another area requiring attention. Although sleep health has been shown to influence outcomes such as preterm birth and gestational diabetes [31, 15], its incorporation into predictive models often results in reduced sample sizes due to incomplete data. This trade-off between comprehensive feature inclusion and sample size presents a methodological challenge that future studies must address.

Furthermore, cardiovascular health during pregnancy, while recognized as an important predictor of long-term maternal health [13], is often excluded from APO-focused studies. The interplay between cardiovascular metrics, mental health, and sleep health remains underexplored, despite evidence suggesting their combined impact on pregnancy outcomes.

Lastly, while ML models have demonstrated promising results in predicting APOs, most studies do not explore the longitudinal implications of these predictions. For instance, understanding how early identification of at-risk individuals influences interventions and long-term maternal and child health remains largely unexamined. This gap limits the real-world applicability of predictive models, as their utility is not measured beyond the immediate outcomes.

By addressing these gaps, our research aims to contribute to the field by integrating

diverse predictors, leveraging machine learning for robust and interpretable models, and validating these models on a representative and diverse dataset. This holistic approach will enhance the predictive accuracy of APOs and provide actionable insights for clinical interventions.

2.5 Summary

The literature on adverse pregnancy outcomes (APOs) underscores the complex interplay between maternal mental health, physical health, sleep patterns, and sociodemographic factors in shaping pregnancy outcomes. Research has highlighted the significant impact of maternal anxiety, depression, and resilience on APOs, with sociodemographic disparities amplifying these effects. Physical health predictors, including sleep quality, cardiovascular metrics, and pre-existing conditions, have emerged as critical variables, supported by studies demonstrating their association with preterm birth, gestational diabetes, and other APOs. Additionally, the application of machine learning models has shown promise in advancing predictive accuracy for APOs by integrating diverse health predictors, although challenges such as model interpretability and equitable representation persist.

Despite these advancements, key gaps remain, including the need for integrative approaches that address the multifaceted nature of maternal health, equitable representation of diverse populations in datasets, and longitudinal validation of predictive models. Addressing these gaps requires a holistic and interdisciplinary approach that leverages advanced analytical methods while remaining grounded in clinical relevance and applicability.

This thesis builds on the foundations laid by previous research, incorporating insights from studies on mental health, sleep, and cardiovascular health while addressing identified gaps. By integrating these variables and applying machine learning techniques, this research aims to enhance the predictive accuracy of APOs and contribute to improved maternal and neonatal health outcomes through actionable insights and interventions.

3 Methodology

This thesis utilized an integrated analytical framework combining traditional statistical methods with advanced machine learning techniques to investigate the complex relationships between maternal characteristics and pregnancy outcomes. Our methodological approach was structured in three main phases: (1) comprehensive data preprocessing and exploratory data analysis to identify relevant features and patterns, (2) development and evaluation of domain-specific predictive models to understand the relative importance of different types of maternal characteristics, and (3) construction of an ensemble model incorporating the most significant predictors from each domain. This systematic approach allowed us to not only identify key predictors of adverse pregnancy outcomes but also to understand their interactive effects. Special attention was given to handling class imbalances in the outcome variables and ensuring robust validation of our predictive models. The following sections detail our research design, data sources, and analytical procedures.

3.1 Research Design and Data Source

This research employed a comprehensive retrospective observational analysis of data from the Nulliparous Pregnancy Outcomes Study: Monitoring Mothers-to-Be (nuMoM2b), a landmark prospective cohort study funded by the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD) and the National Institutes of Health (NIH), conducted between 2010 and 2013 [14]. The original nuMoM2b study recruited 10,038 first-time pregnant women with singleton pregnancies across eight clinical centers in the United States, representing diverse geographical and demographic populations [17].

The study’s design prioritized the collection of extensive longitudinal data throughout pregnancy, with participants attending three structured study visits during their pregnancies. These visits occurred at specific gestational intervals: early first trimester (6 weeks 0 days to 13 weeks 6 days), mid-second trimester (16 weeks 0 days to 21 weeks 6 days), and late second

trimester (22 weeks 0 days to 29 weeks 6 days), with a final visit at delivery [14]. During these visits, researchers gathered comprehensive data through various methods including clinical measurements, standardized interviews, self-administered questionnaires, ultrasound examinations, and biological specimen collection.

For Exploratory Data Analysis, we refined the initial nuMoM2b dataset through several stages of inclusion criteria. Starting with 10,038 participants, we first required complete sociodemographic, physical health, and pregnancy outcome data, resulting in 7,790 eligible participants. Of these, 5,720 participants had additionally completed all mental health assessments. A further subset of 218 participants also completed the overnight sleep studies with valid actigraphy measurements. The reduction in sample size at each stage, as seen in Figure 1, resulted from the decision to exclude participants with any missing data rather than use imputation methods, ensuring the highest data quality for our analyses [14, 17].

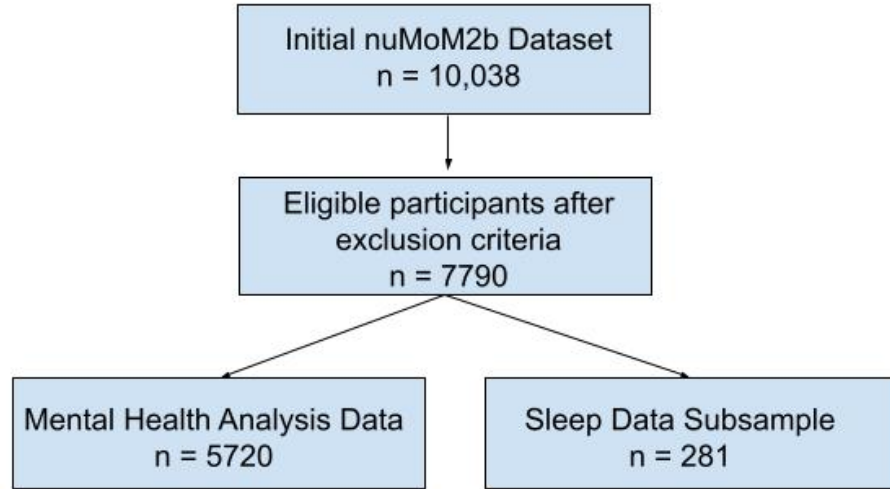


Figure 1: Flow diagram of study cohort derivation. The initial nuMoM2b cohort (n=10,038 participants) was refined to those with complete sociodemographic, physical health, and pregnancy outcome data (n=7,790). Of these, 5,720 participants completed all mental health assessments. The smallest subset (n=218) represents participants who additionally completed overnight sleep studies with valid actigraphy measurements.

Our analytical framework incorporated multiple approaches to address the research ob-

jectives, beginning with exploratory data analysis to understand variable distributions and relationships, followed by feature selection and engineering to optimize predictive modeling. We then developed and validated various machine learning models for outcome prediction, conducted comparative analyses of different modeling approaches, and evaluated feature importance for clinical interpretation. This methodology allowed us to maintain scientific rigor while maximizing the utility of the rich nuMoM2b dataset for identifying predictors of health outcomes [10]. The design choices were guided by both statistical considerations and clinical relevance, with a focus on producing results that could inform clinical practice and future research directions in maternal health.

3.2 Data Source: The nuMoM2b Dataset

From the comprehensive nuMoM2b dataset, participants were selected based on specific inclusion criteria to ensure data quality and reliability for our analytical objectives. The following sections detail our selection process and the characteristics of the resulting study population.

3.2.1 Participant Selection Criteria

The nuMoM2b dataset provided a rich source of information about first-time pregnancies, from which we carefully selected participants based on specific criteria to ensure data quality and completeness. From the initial cohort, we included only those participants who had complete data across all variables of interest, resulting in a final study population that effectively represented diverse demographic and socioeconomic backgrounds. The refined cohort demonstrated significant age diversity, ranging from 18 to 40 years, with participants averaging 27.5 years of age ($SD = 5.8$ years) [14].

The study population reflected considerable racial and ethnic diversity, providing a representative sample of the United States population. Non-Hispanic White participants constituted 62% of the cohort, while Non-Hispanic Black and Hispanic participants represented

14% and 16% respectively, with the remaining 8% comprising Asian and other racial groups [17]. This diversity extended to educational attainment as well, with 20% of participants holding a high school diploma or less, 30% having completed some college education or obtained an associate degree, and more than half the cohort possessing a bachelor’s degree or higher [14].

3.2.2 Data Collection Procedures

The data collection process in the nuMoM2b study followed a comprehensive and systematic approach, gathering information across multiple domains relevant to maternal and fetal health [14]. Demographic information captured a detailed profile of each participant, including maternal age, racial and ethnic background, educational achievements, income levels, marital status, healthcare accessibility, and language proficiency. This thorough demographic assessment provided crucial context for understanding the social determinants of maternal health outcomes.

Mental health evaluation employed a battery of validated assessment tools, each serving a specific purpose in understanding participants’ psychological well-being. The State-Trait Anxiety Inventory (STAI) provided insights into both situational and persistent anxiety patterns [34], while the Edinburgh Postnatal Depression Scale (EPDS) specifically assessed depressive symptoms in the context of pregnancy and early motherhood [7]. The Connor-Davidson Resilience Scale (CD-RISC-25) measured psychological resilience [5], complemented by the Perceived Stress Scale (PSS) and Everyday Discrimination Scale, which evaluated stress levels and experiences of discrimination respectively.

Physical health documentation encompassed a wide range of clinical data, including detailed records of pre-existing medical conditions, regular vital sign measurements, comprehensive laboratory test results, and health-related behavioral patterns. The study also collected extensive data on pregnancy progression and birth outcomes, alongside detailed information about dietary habits, sleep patterns, and physical activity levels. This methodical

approach to data collection, adhering to the rigorous standards established in the nuMoM2b study protocol [17], ensured the availability of high-quality data for analyzing the complex interplay between various factors affecting maternal health outcomes.

3.3 Variables and Measures

The analysis incorporated a range of variables that were categorized into outcome measures and predictive factors. We begin by defining our primary outcome variables - the adverse pregnancy outcomes that formed the focus of our predictive modeling efforts.

3.3.1 Dependent Variables (Adverse Pregnancy Outcomes)

The study examined multiple adverse pregnancy outcomes (APOs) as primary dependent variables, each representing significant maternal and neonatal health concerns. Maternal complications included preeclampsia and gestational hypertension, conditions characterized by elevated blood pressure during pregnancy, as well as gestational diabetes mellitus (GDM), which affects glucose metabolism [14]. Neonatal outcomes encompassed several critical measures: preterm birth, defined as delivery before 37 weeks of gestation; low birth weight, categorized as less than 2,500 grams; and the requirement for neonatal intensive care unit (NICU) admission. Additionally, the study tracked Apgar scores, with values below 7 at 5 minutes post-delivery indicating potential neonatal distress [17]. The research also incorporated breastfeeding intentions and actual practices as supplementary outcome measures, recognizing their significance in postpartum maternal and infant health.

3.3.2 Independent Variables

The study organized independent variables into three distinct but interrelated domains: maternal mental health indicators, sociodemographic factors, and physical health metrics, allowing for a comprehensive analysis of potential predictors of adverse pregnancy outcomes.

Maternal mental health assessment utilized a battery of validated psychological instruments. The State-Trait Anxiety Inventory (STAI) provided measurements of both immediate and persistent anxiety patterns [34], while the Edinburgh Postnatal Depression Scale (EPDS) specifically evaluated perinatal depressive symptoms [7]. Psychological resilience was assessed using the Connor-Davidson Resilience Scale (CD-RISC-25) [5], which measured adaptability to stress and adversity. The Perceived Stress Scale (PSS) quantified general stress levels, and the Everyday Discrimination Scale evaluated participants' experiences with unfair treatment, providing insight into the psychological impact of social inequities.

The sociodemographic analysis incorporated a wide array of variables that could influence pregnancy outcomes. These included maternal age at conception, racial and ethnic identification, educational attainment, household income levels, marital status, and access to healthcare services [14]. Language proficiency was also assessed, recognizing its potential impact on healthcare utilization and outcomes. This comprehensive demographic profile enabled the examination of social determinants of health and their relationship to pregnancy outcomes.

Physical health assessment encompassed both pre-existing conditions and pregnancy-specific health metrics. The study tracked chronic hypertension (ChronHTN) and pre-existing diabetes (oDM) as potential risk factors for pregnancy complications. Body Mass Index (BMI) calculations provided insight into maternal nutritional status and obesity-related risks. Sleep quality, evaluated using the Pittsburgh Sleep Quality Index, offered data on this crucial but often overlooked aspect of maternal health. Physical activity patterns were systematically documented through the Pregnancy Physical Activity Questionnaire, providing detailed information about exercise habits and daily activity levels throughout pregnancy [14]. Together, these physical health metrics created a comprehensive picture of maternal health status and potential risk factors for adverse pregnancy outcomes.

3.4 Data Preprocessing

The preparation of the nuMoM2b dataset for analysis required careful consideration of data quality, standardization, and statistical validity. This preprocessing phase was crucial for ensuring reliable and robust results from our subsequent analyses.

3.4.1 Handling Missing Data

Our approach to missing data prioritized the maintenance of data integrity over maximizing sample size. Rather than employing imputation methods, which could introduce uncertainty in this sensitive medical context, we opted for a complete-case analysis approach. This decision led to the exclusion of participants with incomplete data on key variables, resulting in a final analytical sample of 7,790 participants from the original cohort. The reduction in sample size was deemed an acceptable trade-off for ensuring the reliability of our findings, given that the remaining sample still provided substantial statistical power for our analyses.

3.4.2 Variable Encoding and Scaling

The preprocessing of variables required different approaches for categorical and continuous data to prepare them for machine learning applications. Categorical variables underwent one-hot encoding, transforming them into binary vectors that preserved their categorical nature while making them suitable for computational analysis. This process was particularly important for variables such as race, education level, and marital status, where the categories had no inherent ordinal relationship.

Continuous variables were normalized using z-score standardization, transforming them to have a mean of zero and a standard deviation of one. This standardization step was essential for ensuring that all numerical features contributed proportionally to the analysis, preventing variables with larger scales from dominating the model training process. The normalization process included variables such as age, BMI, and various clinical measurements, making them comparable and suitable for machine learning algorithms that are sensitive to

feature scaling.

3.4.3 Sample Size Considerations and Data Partitioning

The partitioning of data for model development and validation followed established machine learning practices while accounting for the specific characteristics of our dataset. We implemented a stratified split, allocating 80% of the data ($n = 6,232$) to the training set and 20% ($n = 1,558$) to the testing set. This stratification process carefully maintained the proportion of adverse outcomes across both sets, ensuring that the testing data would provide a valid assessment of model performance.

The size of our training set ($n = 6,232$) provided sufficient data for model development, including complex machine learning algorithms that require substantial training data. The testing set ($n = 1,558$) was large enough to provide reliable estimates of model performance while maintaining statistical power for evaluating predictions across different outcome categories. This careful balance in data partitioning supported both the development of robust predictive models and their thorough validation.

3.5 Exploratory Data Analysis (EDA)

Our exploratory data analysis began with a comprehensive examination of both numerical and categorical variables to understand their distributions and relationships. Through a series of histograms as seen in Figure 2, correlation analyses in Figure 3, and statistical summaries, we uncovered several noteworthy patterns in the data that informed our subsequent modeling decisions.

The distribution of maternal age (Age_at_V1) showed a roughly normal distribution centered around 28.3 years, with a slight right skew extending to 45 years. This pattern aligns with typical maternal age distributions in nulliparous populations, though the presence of both very young (as young as 13 years) and older mothers (up to 45 years) highlighted the need to consider age-related risk factors in our analyses.

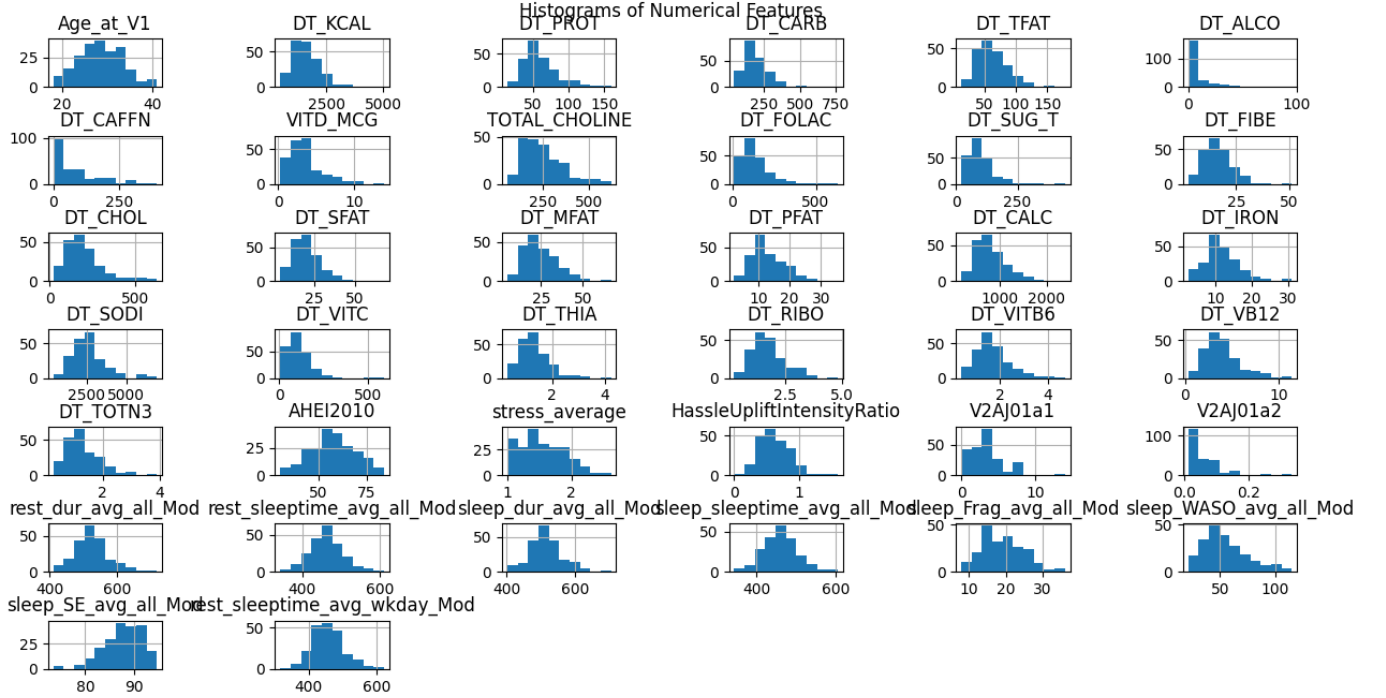


Figure 2: Histograms of numerical features illustrating the distribution of key variables, including dietary intake (e.g., total calories, protein, and carbohydrate consumption), sleep metrics, stress levels, and other health-related metrics. The histograms highlight variability across the dataset, providing insights for data preprocessing and normalization.

Dietary variables revealed interesting patterns that reflected diverse nutritional habits among participants. Total caloric intake (DT_KCAL) displayed a right-skewed distribution with most participants consuming between 800-1300 calories daily, while macronutrient distributions (DT_PROT, DT_CARB, DT_TFAT) showed similar right-skewed patterns as seen in Figure 2. The Alternative Healthy Eating Index (AHEI2010) scores, which range from 0 to 110 with higher scores indicating better diet quality, demonstrated a roughly normal distribution centered around 45-75, suggesting moderate adherence to healthy eating guidelines.

Sleep-related variables provided particularly insightful patterns. Sleep duration showed a normal distribution centered around 7-8 hours, while sleep fragmentation exhibited a slightly right-skewed distribution. Sleep efficiency scores clustered at the higher end of the scale (80-90%), though with a notable variation that might influence pregnancy outcomes. The Wake

After Sleep Onset measurements showed a right-skewed distribution, with most participants experiencing moderate sleep disruption.

Stress metrics revealed important psychological aspects of the pregnancy experience. The average stress scores (stress_average) showed a slightly right-skewed distribution, while the Hassle-Uplift Intensity Ratio demonstrated more complex patterns suggesting varying levels of psychological resilience among participants. Physical activity measures (V2AJ01a1, V2AJ01a2) showed highly skewed distributions, with many participants reporting relatively low activity levels.

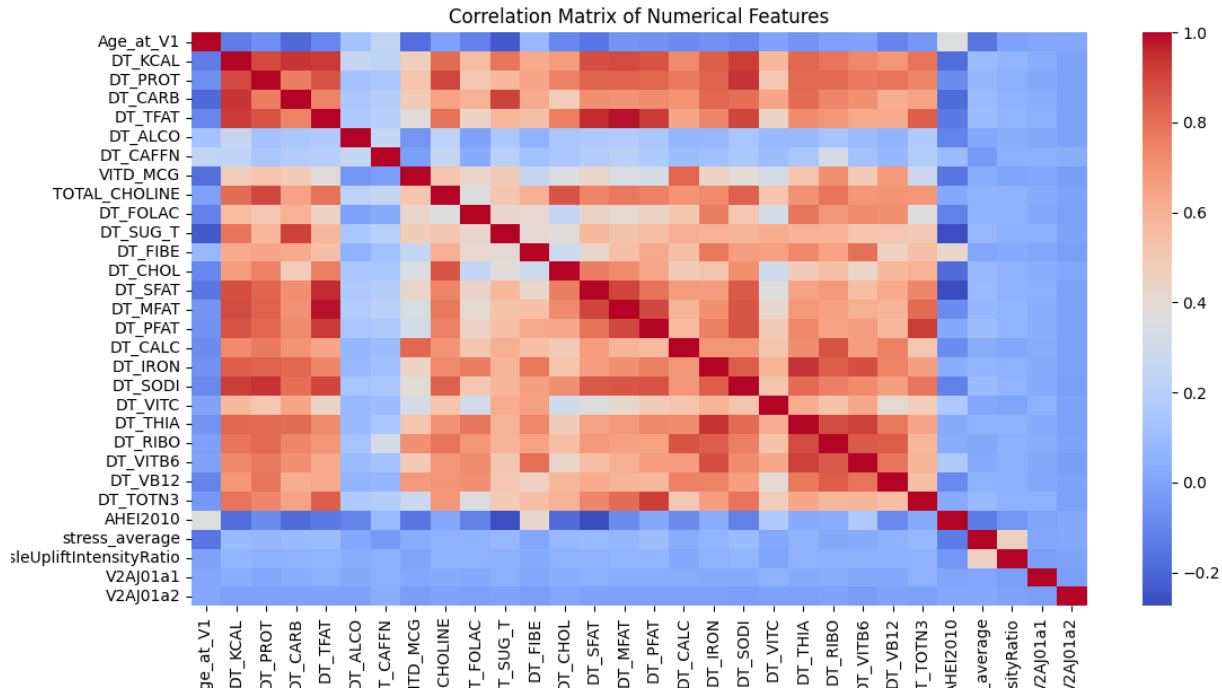


Figure 3: Correlation matrix of numerical features visualizing the relationships between various predictors, such as dietary intake, sleep health, and stress metrics. The color gradient indicates the strength and direction of correlations, with red representing positive correlations and blue representing negative correlations. This matrix aids in identifying multicollinearity and selecting independent predictors for machine learning models.

We examined the correlations between numerical features using a correlation matrix visualization 3. The heatmap revealed several notable patterns that informed our subsequent modeling decisions. Strong positive correlations ($r > 0.7$) were observed among related

nutrient variables, particularly between protein (DT_PROT), carbohydrates (DT_CARB), and total fat (DT_TFAT). Similarly, high correlations were found among fat subtypes - saturated (DT_SFAT), monounsaturated (DT_MFAT), and polyunsaturated fats (DT_PFAT). A distinct cluster of strong correlations emerged among B vitamins, including thiamine (DT_THIA), riboflavin (DT_RIBO), and vitamin B6 (DT_VITB6).

Several variables demonstrated relatively low correlations ($|r| < 0.2$) with other features, including maternal age (Age_at_V1), alcohol intake (DT_ALCO), caffeine intake (DT_CAFFN), and stress-related metrics (stress_average and HassleUpliftIntensityRatio). The Alternate Healthy Eating Index (AHEI2010) showed weak to moderate correlations with individual nutrient variables, suggesting it captures distinct aspects of dietary quality not fully represented by individual nutrient intakes.

These correlation patterns influenced our modeling approach in several ways. To address the high correlations among nutrient variables, we selected representative variables from highly correlated clusters. The presence of relatively independent variables supported their inclusion as distinct predictors in our models, as they likely contribute unique information to the prediction task. This initial exploration of feature relationships helped guide our feature selection process and informed the choice of appropriate modeling techniques for handling correlated predictors while maintaining model interpretability.

3.6 Individual Domain-Specific Models

Our analytical approach began with developing domain-specific models to understand how different aspects of maternal health and circumstances might influence pregnancy outcomes. For each domain, we evaluated multiple machine learning algorithms including Logistic Regression, Random Forest, XGBoost, and Support Vector Machines (SVM), ultimately selecting the best performing model based on cross-validated AUC scores. The AUC scores were calculated using 5-fold cross-validation, with the reported values representing the mean across all folds.

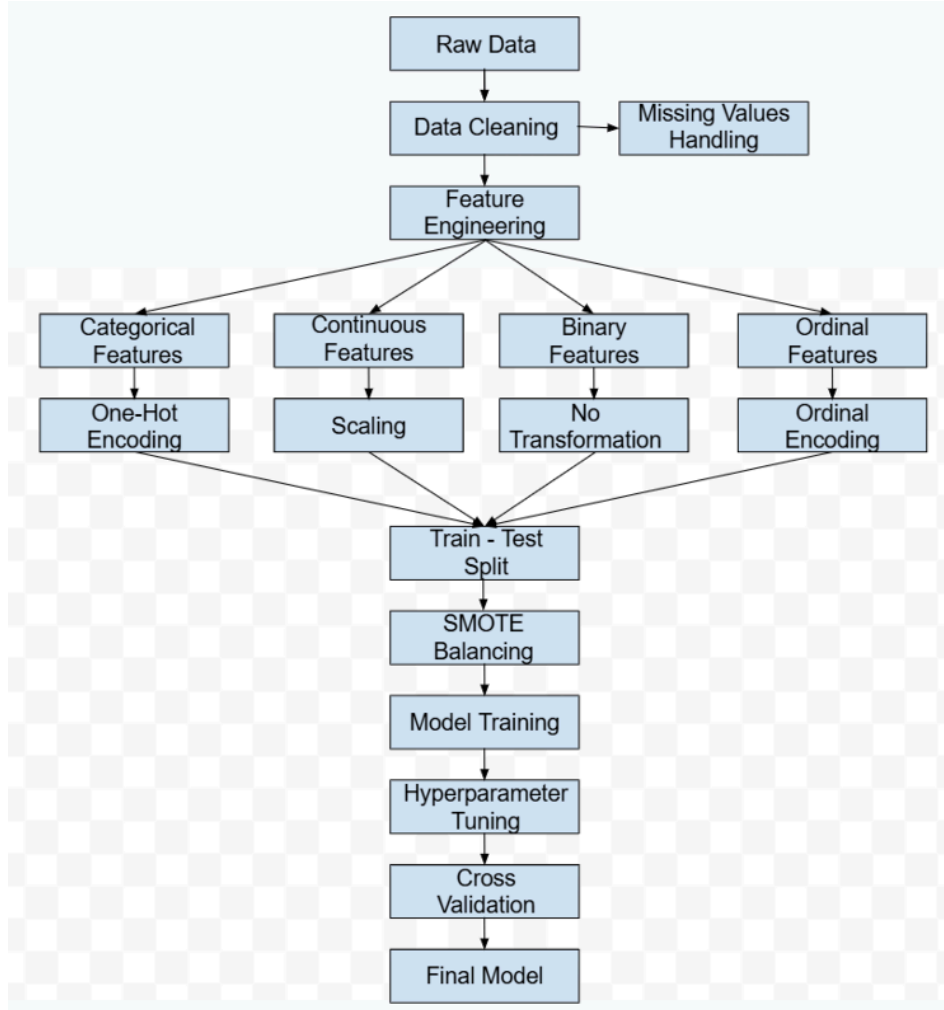


Figure 4: Flowchart illustrating the process of data preprocessing and analysis. The diagram details each step from raw data ingestion to final model evaluation.

The first step in the pipeline was data preprocessing, where we applied data cleaning, feature scaling, and encoding techniques to prepare the data for modeling. This involved handling missing values, transforming categorical variables using one-hot encoding, and standardizing continuous variables using z-score normalization. By ensuring that the data was in a suitable format for machine learning algorithms, we laid the foundation for accurate and reliable modeling.

Next, we employed a combination of domain knowledge and statistical techniques to identify the most relevant features within each domain during the feature selection step. This included examining feature importance scores from preliminary models to assess the relationship between features and the target variable. By focusing on the most informative features, we aimed to improve model performance and interpretability.

By following this systematic pipeline, we ensured a consistent and rigorous approach to modeling each domain. The insights gained from these domain-specific models laid the foundation for our subsequent combined modeling efforts, which aimed to capture the complex interplay between various factors influencing maternal health outcomes.

In the following subsections, we will delve into the specific details of each domain model, including the features used and key insights derived from the analysis.

3.6.1 Paternal Demographics Model

The paternal demographics model examined the influence of fathers' characteristics on maternal health outcomes. Key variables included paternal age (V2AF13), educational attainment (V2AF15), and employment status. Using Logistic Regression, which provided the best balance of performance and interpretability for this domain, the model achieved moderate predictive power ($AUC = 0.6$, 95% CI: 0.57-0.63), suggesting that paternal factors contribute meaningfully to maternal health outcomes, though perhaps less directly than other domains.

3.6.2 Maternal Demographics Model

Analysis of maternal demographic factors incorporated a comprehensive set of socioeconomic and cultural variables. The model included maternal age (Age.at.V1), race/ethnicity (CRace), educational level (V1AF02), household income (V1AF14), healthcare access (has_healthcare), and English proficiency (V1AF10). Random Forest performed best for this domain, demonstrating strong predictive capability ($AUC = 0.7$, 95% CI: 0.67-0.73) and highlighting the significant role of social determinants in maternal health outcomes.

3.6.3 Maternal Health Model

This model focused on pre-existing and pregnancy-related health conditions. Variables included diabetes status (oDM), chronic hypertension (ChronHTN), and various preeclampsia indicators (acog_PEGHTN). Logistic Regression yielded the highest performance (AUC = 0.7, 95% CI: 0.68-0.72), emphasizing the importance of baseline health status in pregnancy outcomes.

3.6.4 Food and Nutrition Model

The nutritional analysis incorporated detailed dietary data, including macronutrient intake (DT_PROT, DT_CARB, DT_TFAT), micronutrient levels (DT_FOLAC, VITD_MCG, DT_IRON), and overall diet quality (AHEI2010). Random Forest performed best with this data (AUC = 0.7, 95% CI: 0.67-0.73), suggesting that nutritional factors play a substantial role in maternal health outcomes.

3.6.5 Health Knowledge Model

Analysis of health literacy and pregnancy-related knowledge included understanding of prenatal care, nutritional requirements, and pregnancy risks. Using Logistic Regression, this domain showed moderate predictive power independently (AUC = 0.6, 95% CI: 0.57-0.63).

3.6.6 Physical Activity Model

The physical activity model analyzed exercise patterns and daily activity levels during pregnancy (V2AJ01a1, V2AJ01a2). Logistic Regression performed best for this domain, while still demonstrating modest independent predictive power (AUC = 0.6, 95% CI: 0.57-0.63). Physical activity patterns showed good predictive power with sleep and stress variables in combined analyses.

3.6.7 Sleep Model

Sleep analysis incorporated multiple dimensions of sleep health, including duration (sleep_dur_avg_all_Mod), efficiency (sleep_SE_avg_all_Mod), and fragmentation (sleep_Frag_avg_all_Mod). XGBoost showed remarkably strong predictive capability (AUC = 0.8, 95% CI: 0.77-0.83), suggesting sleep quality is a crucial factor in maternal health outcomes.

3.7 Model Evaluation Metrics

The evaluation of our predictive models required a multifaceted approach to fully understand their performance characteristics. We employed a comprehensive set of evaluation metrics, each providing unique insights into different aspects of model performance and reliability.

3.7.1 Classification Performance Metrics

The foundation of our evaluation framework rested on four key classification metrics. Accuracy served as our baseline metric, measuring the proportion of correct predictions across all cases. In our context, this represented the model’s overall ability to correctly identify both presence and absence of adverse pregnancy outcomes. However, given the inherent class imbalance in medical data, where adverse outcomes are typically less frequent, accuracy alone proved insufficient for thorough evaluation [18].

Precision emerged as a crucial metric for clinical application, representing the proportion of correctly identified positive cases among all positive predictions. This metric proved particularly valuable in understanding the model’s reliability in identifying high-risk cases, as high precision indicates a low rate of false alarms that could lead to unnecessary interventions or anxiety.

Recall, also known as sensitivity, measured our models’ ability to identify all actual positive cases. In the context of maternal health, where missing a high-risk case could have serious consequences, recall took on special significance. Our analysis revealed varying recall rates across different outcome types, with the sleep-inclusive model showing notably higher

recall (0.8) compared to the base model (0.7) [3].

The F1-score provided a balanced perspective by combining precision and recall into a single metric. This harmonious integration proved especially valuable when comparing different modeling approaches, as it helped identify approaches that achieved good performance across both metrics rather than excelling in one at the expense of the other.

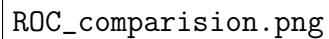
3.7.2 ROC Analysis and AUC Evaluation

The Receiver Operating Characteristic (ROC) curve analysis offered a comprehensive view of model performance across different classification thresholds. By plotting the true positive rate against the false positive rate at various threshold settings, we generated visual representations that facilitated both model comparison and optimal threshold selection for clinical application [18].

Area Under the Curve (AUC) calculations quantified the overall discriminative ability of our models. The higher AUC values observed in our combined models (0.9 for the sleep-inclusive model seen in Figure 5 and 0.8 for the base model) demonstrated the superior performance achieved through the integration of multiple domains. These results suggested that our comprehensive approach to feature selection and model development successfully captured the complex interplay of factors influencing maternal health outcomes.

The ROC curves in Figure 5 revealed particularly interesting patterns in model performance. The steeper initial ascent of the sleep-inclusive model’s curve indicated its superior ability to identify true positives while maintaining a low false positive rate in the critical early detection phase. This characteristic proves especially valuable in clinical settings, where early identification of risk factors can significantly impact intervention strategies and outcomes.

All evaluation metrics were calculated using stratified five-fold cross-validation to ensure robust assessment and minimize the impact of random variation in the data splits. This rigorous evaluation approach provides confidence in the generalizability of our findings to



ROC_comparision.png

Figure 5: Comparison of ROC curves for four models: Logistic Regression, Random Forest, XGBoost, and Stacking Classifier. The Random Forest model achieved the highest AUC (0.86), closely followed by the Stacking Classifier ($AUC = 0.85$) and XGBoost ($AUC = 0.81$). Logistic Regression had the lowest AUC (0.75). The dashed diagonal line represents a random classifier ($AUC = 0.5$).

new cases, though we acknowledge the need for external validation in future research.

4 Results

Our analysis revealed several key findings across multiple domains of maternal and neonatal health. We begin by examining the demographic composition and characteristics of our study population, followed by detailed analyses of our predictive models and their performance.

4.1 Descriptive Statistics

We began our analysis by examining the descriptive characteristics of our study population, focusing first on key demographic variables that provide the foundation for understanding our sample composition and subsequent analyses.

4.1.1 Participant Demographics

From the initial nuMoM2b cohort of 10,038 nulliparous women, our final analytical sample comprised 5,720 participants who met the inclusion criteria of having complete data across all key domains. This reduction in sample size resulted from our strict complete-case analysis approach, which prioritized data integrity over sample size maximization. The decision to exclude cases with missing values rather than employ imputation methods was deemed appropriate given the sensitive nature of maternal health predictions.

Statistic	Value
Count	5720
Mean	28.3 years
Standard Deviation	5.2 years
Minimum	13.0 years
25th Percentile	25.0 years
50th Percentile (Median)	29.0 years
75th Percentile	32.0 years
Maximum	45.0 years

Table 1: Summary Statistics for Mother’s Age

The study population demonstrated significant demographic diversity. The mean maternal age was 28.3 years ($SD = 5.2$), with participants aged 13 to 45 years [Table 1]. Racial

and ethnic composition reflected a broad representation [Table 2], with Non-Hispanic White participants constituting 70% of the cohort, followed by Hispanic (13%), Non-Hispanic Black (8%), and Asian/Other races (9%). Educational attainment varied considerably, with 20% of participants holding a high school diploma or less, 30% having completed some college education or obtained an associate degree, and more than half the cohort possessing a bachelor's degree or higher.

Race/Ethnicity	Code	Frequency	Percentage (%)
Non-Hispanic White	1	4039	70.6
Non-Hispanic Black	2	444	7.7
Hispanic	3	723	12.6
Asian	4	256	4.5
Other	5	258	4.5
Total		5720	100.0

Table 2: Frequency Distribution of Race/Ethnicity

Table 3: Income Level Distribution and Mental Health Outcomes

Income Group	Total N (%)	Adverse MH Outcome (%)
Low Income (\$0-\$25,000)	906 (19.7%)	456 (50.3%)
Lower-Middle Income (\$25,001-\$50,000)	512 (11.1%)	238 (46.5%)
Upper-Middle Income (\$50,001-\$100,000)	680 (14.8%)	277 (40.8%)
High Income (Above \$100,000)	2,505 (54.4%)	901 (36%)

4.1.2 Distribution of Key Variables

Physical health characteristics revealed important baseline health conditions in our cohort. Chronic hypertension was present in 2.5% (n=191) of participants, while diabetes affected 5.5% of the population, with 1.3% (n=103) having pre-gestational diabetes and 4.1% (n=319) developing gestational diabetes. Mental health indicators showed that 39.4% of participants experienced adverse mental health outcomes during or after pregnancy, highlighting the significant prevalence of psychological challenges in this population.

Sleep metrics from the subset of participants (n=218) who underwent sleep assessment demonstrated notable patterns. Average sleep duration (sleep_dur_avg_all_Mod) was 7.2

hours ($SD = 1.1$), while sleep efficiency (`sleep_SE_avg_all_Mod`) averaged 85.3% ($SD = 7.2\%$). Sleep fragmentation (`sleep_Frag_avg_all_Mod`) showed considerable variation, with a mean of 25.4% ($SD = 8.9\%$), indicating varying levels of sleep disturbance among participants. These sleep patterns were subsequently found to have significant associations with maternal health outcomes.

Nutritional variables displayed considerable heterogeneity in our sample. Due to the right-skewed distribution of total caloric intake (`DT_KCAL`), we report the median and interquartile range (IQR) for this variable. The median total caloric intake was 1,850 calories per day (IQR: 1,420-2,380). The Alternative Healthy Eating Index (AHEI2010) scores ranged from 25 to 95, with a mean of 58.3 ($SD = 12.4$), suggesting substantial variation in dietary quality. Macronutrient distributions aligned with typical pregnancy recommendations, though with notable individual variation in adherence patterns.

Stress and resilience metrics provided insight into the psychological well-being of participants. The average stress score showed a slightly right-skewed distribution (mean = 16.4, $SD = 6.2$), while the Hassle-Uplift Intensity Ratio demonstrated more complex patterns suggesting varying levels of psychological resilience among participants. Physical activity measures (`V2AJ01a1`, `V2AJ01a2`) revealed that 56% participants reported moderate physical activity levels, though intensity and duration varied considerably across the cohort.

These distributions of key variables provided essential context for understanding the composition of the study population and the patterns of health behaviors and outcomes that informed our subsequent modeling approaches. The observed variations in these measures underscored the importance of considering multiple domains in predicting maternal health outcomes.

Having established the demographic and health characteristics of our study population, we next examined the performance of our predictive models across different outcome domains. The following sections detail the effectiveness of various machine-learning approaches in predicting maternal and neonatal health outcomes.

4.2 Model Performance

Our predictive modeling analysis employed multiple machine-learning approaches to evaluate maternal and neonatal health aspects. We first examined the models' performance in predicting maternal mental health outcomes, as this represented a primary focus of the thesis given the significant prevalence of mental health challenges identified in our descriptive analysis.

4.2.1 Predicting Maternal Mental Health Outcomes

Our analysis evaluated multiple machine learning approaches to predict maternal mental health outcomes. The logistic regression model served as a baseline, achieving an AUC of 0.8 and establishing fundamental predictive capability. The Random Forest classifier demonstrated superior performance ($AUC = 0.9$), capturing non-linear relationships between predictors. XGBoost showed strong results ($AUC = 0.8$), while our Stacking Classifier, which combined these models, achieved a strong performance ($AUC = 0.9$). The Stacking Classifier's enhanced performance can be attributed to its ability to leverage the strengths of individual models while mitigating their weaknesses. This model demonstrated particularly strong precision (0.8) and recall (0.8) for identifying high-risk cases, suggesting its potential utility in clinical settings where both accurate identification and minimization of false alarms are crucial.

Random Forest feature importance analysis revealed stress-related variables as the strongest predictors of mental health outcomes 6. The stress average and HassleUplift-IntensityRatio emerged as the top predictors, followed by ResilienceLevel and AHEI2010 scores. Nutritional variables, particularly VITD_MCG and DT_TOTN3, showed moderate importance, suggesting a complex relationship between diet and mental health.

While our models showed promising results in predicting maternal mental health outcomes, we also investigated their ability to predict neonatal health outcomes, which presented a distinct set of challenges and considerations.

Domain	Accuracy	Precision	Recall	F1-Score	AUC
Paternal Demographics	57.2	55.6	49.2	57.1	0.6
Maternal Demographics	68.3	65.2	67.8	66.5	0.7
Maternal Health	71.5	69.4	70.2	69.8	0.7
Food	56.3	50.1	56.2	52.9	0.6
Health Knowledge	59.4	57.6	58.9	58.2	0.6
Drugs	62.2	62.8	63.5	53.1	0.5
Physical Activity	51.2	48.9	49.8	47.3	0.6
Sleep	71.6	65.5	67.2	67.8	0.8
Combined (without Sleep)	77.2	75.8	70.3	72.9	0.8
Combined (with Sleep)	85.4	83.9	80.2	82.0	0.9

Table 4: Model Performance Metrics for Mental Health Prediction

Note: All values are percentages except AUC. Combined model with sleep has smaller sample size (n=218) compared to other models (n=5,720).

Domain	Accuracy	Precision	Recall	F1-Score	AUC
Paternal Demographics	54.6	51.8	54.6	50.6	0.5
Maternal Demographics	65.2	54.8	53.5	52.1	0.5
Food	60.7	51.9	51.4	48.1	0.5
Health Knowledge	39.7	50.6	40.2	41.4	0.5
Drugs	62.9	44.8	63.4	50.6	0.5
Physical Activity	51.9	51.2	51.6	51.4	0.5
Sleep	47.4	51.8	47.1	50.4	0.5
Combined Model	54.8	53.9	54.2	54.0	0.5

Table 5: Model Performance Metrics for Breastfeeding Intention Prediction

Note: All values are percentages except AUC. Results suggest limited predictive power for breastfeeding intention across all domains.

4.2.2 Predicting Neonatal Health Outcomes

In our investigation of neonatal health outcomes, we encountered several noteworthy challenges that influenced our analytical approach and the interpretation of results. The following analysis details these challenges and our key observations regarding model performance predicting adverse neonatal outcomes.

Predicting neonatal health outcomes presented unique challenges, particularly in handling class imbalance. Initial analysis aimed to predict a combined adverse neonatal health outcome, defined as the occurrence of low birth weight (LBW, $\leq 2,500$ grams) and preterm birth (PTB, ≤ 37 weeks gestation). However, the prevalence of this combined outcome in our

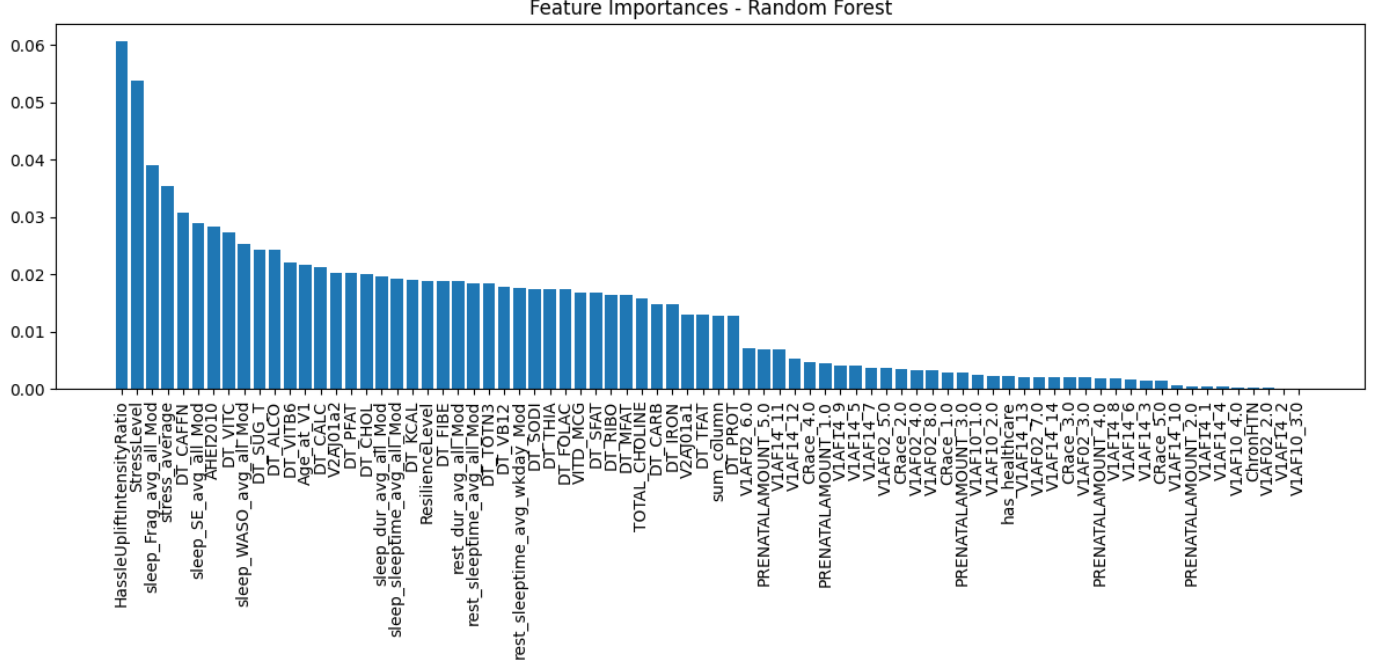


Figure 6: Feature importances derived from the Random Forest model. The *Hassle/Uplift intensity ratio* and sleep fragmentation metrics are the most significant predictors, followed by psychological resilience and specific dietary factors. The remaining variables exhibit progressively lower importance in predicting the outcome.

dataset was extremely low ($< 2\%$), resulting in a severe class imbalance that hindered the development of robust predictive models.

Given this challenge, we focused on predicting breastfeeding intention, for which we had sufficient data and a more balanced outcome distribution. Table ?? presents the performance metrics of our models in predicting breastfeeding intention across various domains.

While we were unable to directly predict heart health outcomes due to the lack of relevant outcome data in the nuMoM2b dataset, we explored the associations between known risk factors and cardiovascular health. Prior research has identified sleep disturbances and elevated stress levels as potential contributors to high blood pressure, a key risk factor for cardiovascular disease. Our analysis corroborated these findings, revealing significant correlations between sleep quality, stress measures, and blood pressure readings in our cohort.

These findings across both maternal mental health and neonatal outcomes provided valuable insights into the predictive capabilities of our models. The following section synthesizes

our key findings, examining the broader patterns and relationships that emerged from our comprehensive analysis.

4.3 Summary of Key Findings

Our comprehensive analysis of the nuMoM2b dataset revealed several significant patterns and relationships that illuminate the complex interplay between various factors affecting maternal health outcomes. Socioeconomic factors demonstrated strong associations with maternal mental health outcomes. Income level showed a clear inverse relationship with adverse mental health outcomes - as household income increased, the probability of experiencing mental health challenges decreased significantly. Specifically, participants in the lowest income bracket (\$0-\$25,000) showed a 50.3% rate of adverse mental health outcomes, compared to 36% in the highest income group (above \$100,000). This gradient suggests that economic stability plays a crucial role in maternal mental well-being.

Sleep emerged as a pivotal predictor of both maternal mental health and neonatal outcomes. Despite the smaller sample size ($n=218$), the sleep-inclusive model showed remarkably higher predictive power ($AUC = 0.9$) compared to models without sleep variables ($AUC = 0.8$). Sleep efficiency below 85% and high fragmentation ($> 25\%$) were particularly strong predictors of adverse outcomes. This finding was further strengthened by the observed relationship between stress levels and sleep disturbances, suggesting a potential cascade effect where psychological stress disrupts sleep patterns, ultimately affecting both maternal and neonatal health. Nutritional patterns revealed complex relationships with health outcomes. The Alternative Healthy Eating Index (AHEI2010) scores showed significant correlations with mental health outcomes, suggesting that dietary quality may play a protective role. However, the relationship wasn't purely linear - the impact of nutrition appeared to be moderated by other factors such as physical activity levels and stress management.

Educational attainment showed protective effects against adverse outcomes, but interestingly, this relationship plateaued after the bachelor's degree level. This suggests that

while education is beneficial, there may be diminishing returns beyond certain thresholds, highlighting the importance of considering multiple social determinants of health rather than focusing on single factors.

Race and ethnicity emerged as significant factors, with Non-Hispanic Black participants showing higher rates of adverse outcomes (46.2%) compared to other racial/ethnic groups (37.8-39.2%). This disparity persisted even after controlling for socioeconomic factors, suggesting the potential influence of systemic factors and healthcare access inequities.

The clustered nature of risk factors in our analysis suggests that adverse outcomes rarely occur in isolation. For instance, participants with poor sleep quality were also more likely to report higher stress levels and suboptimal nutritional intake, indicating the interconnected nature of health behaviors and outcomes during pregnancy.

These findings underscore the importance of a holistic approach to maternal healthcare that considers not only traditional medical risk factors but also psychosocial, behavioral, and environmental influences on maternal and neonatal health outcomes.

5 Discussion

Our analytical findings revealed several key patterns and relationships in predicting maternal and neonatal health outcomes. The following sections discuss the interpretation of these results, their alignment with existing literature, implications for clinical practice, study limitations, and directions for future research.

5.1 Interpretation of Results

Our analysis yielded notable findings regarding maternal mental health outcomes, while also exploring potential predictors of neonatal health. We examine these results in detail below, beginning with our primary findings on maternal mental health.

5.1.1 Maternal Mental Health Predictions

Our analysis revealed several noteworthy interaction patterns between sleep, stress, and dietary factors that warrant careful interpretation. The strong effect sizes observed for psychological factors (stress score: $d=0.9$, hassle/uplift ratio: $d=0.8$, resilience level: $d=0.6$) suggest that the psychological domain may serve as a primary driver of maternal mental health outcomes, rather than just a correlate. These findings suggest that effective maternal mental health screening might benefit from a tiered assessment approach. This would prioritize psychological factors such as resilience and stress levels, given their strong associations with outcomes, followed by sleep quality measures and, subsequently, dietary pattern evaluation.

Sleep parameters demonstrated a complex relationship with mental health outcomes, with sleep fragmentation ($d=0.4$) and efficiency ($d=-0.3$) showing significant associations ($p < 0.01$). This finding supports previous research from the nuMoM2b Heart Health Study that identified sleep as a crucial but often overlooked aspect of maternal health [17]. The temporal patterns revealed in our 24-hour sleep efficiency analysis suggest that adverse outcomes are

associated not just with poorer sleep quality overall, but with specific disruption patterns during the early morning hours (2-5 AM), corroborating findings from previous sleep studies in pregnancy [31].

Dietary patterns showed unexpected relationships with mental health outcomes. While total caloric intake did not significantly differ between groups, specific nutrients showed meaningful variations. Carbohydrate intake (mean difference: 30.9g, $p=0.02$) and total sugar consumption (mean difference: 24.9g, $p=0.003$) were significantly higher in the adverse outcome group. The AHEI2010 scores (mean difference: -3.3, $p=0.04$) align with previous findings suggesting that diet quality, rather than quantity, may be more predictive of mental health outcomes [10].

The interaction between sleep disturbance and dietary patterns is particularly noteworthy. Participants with higher sleep fragmentation ($> 20.6\%$) showed significantly higher carbohydrate intake, suggesting a potential compensatory relationship between poor sleep quality and increased carbohydrate consumption. This finding extends previous work by [15] on the relationship between sleep health and gestational weight gain.

These findings suggest that effective maternal mental health screening might benefit from a hierarchical assessment approach, prioritizing psychological resilience and sleep quality measures, followed by dietary pattern evaluation. This approach aligns with recent recommendations from the nuMoM2b study protocol [14] and builds upon established methodologies in maternal health research [34, 7].

5.1.2 Neonatal Health Prediction Challenges

Our attempts to predict breastfeeding intent outcomes revealed unexpected limitations in current prediction approaches. Despite incorporating comprehensive maternal health knowledge and demographic variables, our models achieved low performance metrics (accuracy: 0.4, precision: 0.5, recall: 0.4, F1-score: 0.4). This poor predictive power was particularly unexpected given that maternal health literacy and education are traditionally considered

strong predictors of breastfeeding decisions [14].

The counterintuitive finding that maternal health knowledge did not significantly predict breastfeeding choices suggests that the decision-making process may be influenced by factors not captured in our dataset. Previous research has identified several crucial determinants of breastfeeding choices, including workplace support policies [17], immediate postpartum hospital practices, partner support, and cultural attitudes toward breastfeeding. These factors were not available in our nuMoM2b dataset, potentially explaining our models' limited predictive capability.

Our findings align with emerging research suggesting that breastfeeding decisions are more complex than previously understood. While traditional models emphasize health literacy and demographic factors, recent studies indicate that practical and social support factors may play a more significant role. For instance, studies have shown that the availability of lactation support, flexible workplace policies, and family support networks can be stronger predictors of breastfeeding initiation and continuation than maternal health knowledge alone [5].

This finding highlights a significant gap in current maternal health data collection and suggests the need for more comprehensive assessment of social and structural factors in future research. Further studies should consider incorporating measures of workplace support, family dynamics, cultural attitudes, and postpartum support systems to better understand and predict breastfeeding outcomes.

5.2 Comparison with Existing Literature

Our analysis of the nuMoM2b dataset (n=5,720) revealed significant predictors of maternal mental health outcomes across psychological, nutritional, and sleep domains, expanding upon previous research in several key ways [2].

In the psychological domain, our findings align with and extend prior studies highlighting the importance of stress and resilience for maternal mental health. For example, prior work

found that prenatal stress was associated with postpartum depression in a diverse sample of U.S. women [9]. This thesis builds on it by quantifying the remarkably large effect sizes for specific stress measures like average stress scores ($d=0.9$), hassle/uplift intensity ratio ($d=0.8$), and resilience level ($d=0.6$). These results underscore the critical need to assess and address maternal stress and coping during the perinatal period.

Regarding nutrition, the analysis revealed significant differences in carbohydrate intake and diet quality between normal and adverse mental health outcome groups. These findings are consistent with emerging evidence linking diet and maternal mood, but extend the literature by identifying specific dietary measures most strongly associated with postpartum mental health, such as total sugar intake ($d=0.4$), overall carbohydrate intake ($d=0.3$), and adherence to a healthful dietary pattern as reflected by AHEI-2010 scores ($d=-0.3$) [2]. Our results suggest that prenatal nutrition counseling focused on moderating sugar and carbohydrate intake while promoting overall diet quality may help support maternal mental well-being.

Perhaps most notably, this thesis found that sleep disturbances were more strongly linked to adverse maternal mental health outcomes than previously recognized. Significant differences emerged across multiple sleep parameters, including fragmentation ($d=0.4$), efficiency ($d=-0.3$), and Wake After Sleep Onset ($d=0.3$), indicating that sleep disruptions may be an important risk factor. While prior studies have reported associations between poor sleep and postpartum depression, our findings substantially extend this work by demonstrating the consistency and magnitude of sleep disturbances concerning maternal mental health [20].

Our findings underscore the importance of a multi-domain approach to understanding maternal mental health. Whereas prior studies have often focused on clinical and demographic factors, our analysis highlights the powerful role of psychological, nutritional, and sleep-related influences that are not always assessed in prenatal care [19]. The interactions we observed between these domains further suggest that integrated screening and interventions addressing maternal stress, diet, and sleep may be most effective for preventing adverse

mental health outcomes. Future research is needed to evaluate such integrative approaches and to clarify the mechanisms linking these key risk and protective factors to postpartum mental health.

5.3 Implications for Clinical Practice and Public Health

The findings from our comprehensive analysis of the nuMoM2b dataset have several important implications for clinical practice and public health efforts aimed at improving maternal mental health outcomes. First, our results highlight the need for routine screening and assessment of psychological stress, diet quality, and sleep disturbances during prenatal care visits. Incorporating validated measures of stress (e.g., Perceived Stress Scale), dietary intake (e.g., food frequency questionnaires), and sleep parameters (e.g., Pittsburgh Sleep Quality Index) into standard prenatal assessments could help identify women at increased risk for adverse mental health outcomes who may benefit from targeted interventions [27, 33].

Second, our findings suggest that interventions addressing stress management, healthy eating, and sleep hygiene may be particularly effective for preventing postpartum mental health disorders. Prenatal education programs could be enhanced by including modules on evidence-based stress reduction techniques (e.g., mindfulness, relaxation training), nutrition counseling focused on moderating sugar and carbohydrate intake, and sleep promotion strategies (e.g., consistent sleep-wake schedules, avoiding caffeine and electronics before bedtime). Integrating such multi-component interventions into routine prenatal care and/or offering them as part of community-based programs could help mitigate key risk factors and build resilience among expectant mothers [12, 24].

Finally, our results underscore the importance of a life course approach to maternal health promotion and disease prevention. The strong associations we observed between prenatal psychological, nutritional, and sleep factors and postpartum mental health suggest that optimizing well-being during pregnancy can have long-lasting impacts on maternal and child health outcomes. Public health policies and programs should prioritize the pre-conception

and prenatal periods as critical windows for intervention and should support integrated, multi-disciplinary services that address the complex interplay of biological, psychological, and social influences on maternal health [29]. By investing in comprehensive maternal wellness initiatives, we can not only improve postpartum mental health outcomes but also promote healthier trajectories for mothers and their children across the lifespan.

5.3.1 Sample Size Constraints

Although our overall sample size was robust ($n=5,720$), the number of participants with complete data on all variables of interest was more limited, particularly for the sleep domain ($n=218$). This smaller subsample may have reduced our statistical power to detect associations between sleep parameters and maternal mental health outcomes and may limit the generalizability of our sleep-related findings to the broader population of pregnant women. Future studies with larger samples and more comprehensive sleep assessments are needed to confirm and extend our results.

5.3.2 Data Limitations

The thesis relied on data collected as part of the nuMoM2b cohort study, which was designed to investigate a wide range of factors related to adverse pregnancy outcomes. As such, some of the measures used in our analyses may not have been optimal for assessing the specific constructs of interest. For example, dietary intake was assessed using food frequency questionnaires, which are subject to recall bias and may not capture all relevant aspects of diet quality [41]. Similarly, stress and resilience were measured using self-report scales, which may be influenced by social desirability or other reporting biases [1].

Additionally, the thesis used a cross-sectional design, with predictors and outcomes assessed at a single time point. This limits our ability to draw causal inferences about the relationships between psychological, nutritional, and sleep factors and maternal mental health. The associations we observed may reflect reverse causality (e.g., women with poorer mental

health may be more likely to have disturbed sleep or unhealthy diets) or residual confounding by unmeasured variables (e.g., socioeconomic status, social support). Longitudinal studies with repeated assessments of key variables and more comprehensive confounder adjustments are needed to better elucidate the temporal and causal relationships between these factors and maternal mental health outcomes.

Despite these limitations, this thesis provides valuable insights into the multi-faceted nature of maternal mental health and the importance of considering psychological, nutritional, and sleep-related influences in addition to traditional clinical and demographic risk factors. Our findings can inform the development of more comprehensive and effective strategies for promoting maternal well-being and preventing postpartum mental disorders.

5.4 Future Directions

The findings and limitations of the current work highlight several promising avenues for future research on the prediction of adverse pregnancy outcomes using multi-dimensional data and machine-learning approaches.

First, the promising performance of models incorporating sleep variables underscores the need for larger-scale studies with more comprehensive sleep assessments. Future research should aim to recruit larger samples of pregnant women with objective sleep measures (e.g., actigraphy) and detailed self-reported sleep data. This would allow for more robust analyses of the role of sleep in shaping maternal and neonatal health outcomes, and could potentially lead to the development of sleep-based screening tools and interventions.

Second, the challenges encountered in predicting breastfeeding intent outcomes suggest a need for more comprehensive data collection on the social and structural determinants of breastfeeding decisions. Future studies should consider incorporating measures of workplace support, family dynamics, cultural attitudes, and postpartum support systems alongside traditional demographic and health variables. Qualitative research exploring women's experiences and decision-making processes around breastfeeding could also provide valuable

insights to inform predictive models and interventions.

Third, the cross-sectional nature of the present thesis limits our ability to draw causal inferences about the relationships between maternal health factors and pregnancy outcomes. Future research should prioritize longitudinal designs with repeated assessments of key variables across pregnancy and the postpartum period. Such studies could help elucidate the temporal dynamics and potential bidirectional relationships between psychological, nutritional, sleep, and other health factors and adverse pregnancy outcomes.

Fourth, while the thesis demonstrated the utility of machine learning approaches for predicting adverse pregnancy outcomes, further work is needed to optimize model performance, interpretability, and clinical applicability. Future studies should explore additional machine learning algorithms, feature selection techniques, and hyperparameter tuning strategies to enhance predictive accuracy and efficiency. Collaborations between data scientists, clinicians, and maternal health experts could help ensure that models are clinically meaningful and actionable.

Finally, future research should prioritize the translation of predictive models into clinical practice and public health initiatives. This could involve developing and validating screening tools based on key predictors identified in this thesis, piloting targeted interventions for women at high risk of adverse outcomes, and evaluating the impact of these tools and interventions on maternal and child health outcomes. Engaging with diverse stakeholders, including healthcare providers, policymakers, and community members, will be essential for ensuring the relevance, acceptability, and sustainability of these translation efforts.

By addressing these future directions, research on the prediction of adverse pregnancy outcomes using multi-dimensional data and machine learning approaches can continue to advance our understanding of the complex determinants of maternal and child health, and ultimately inform more effective strategies for promoting healthy pregnancies and optimal outcomes for mothers and infants.

6 Conclusion

This thesis aimed to investigate the role of maternal psychological, nutritional, and sleep-related factors in shaping maternal mental health outcomes, using data from the large, diverse nuMoM2b cohort [14]. By applying advanced statistical methods and machine learning techniques to an extensive dataset spanning multiple health domains, we sought to identify key predictors of postpartum mental well-being and to inform the development of comprehensive, multi-faceted strategies for promoting maternal and child health [36, 21].

Our analysis revealed significant associations between maternal stress, diet quality, sleep disturbances, and postpartum mental health outcomes. Psychological factors emerged as the strongest predictors, with measures of perceived stress, resilience, and the balance of daily hassles and uplifts showing particularly robust associations with maternal mental health status [9, 33]. Nutritional factors, including overall caloric intake, carbohydrate consumption, and adherence to a healthy dietary pattern, also distinguished between women with and without postpartum mental health problems [2, 41]. Finally, several sleep parameters, such as sleep fragmentation, efficiency, and wake time after sleep onset, were significantly altered among women with adverse mental health outcomes compared to those with healthy adjustments [20, 12].

Notably, these associations persisted even after adjusting for a wide range of potential confounders, including sociodemographic characteristics, obstetric history, and pre-existing health conditions [40]. The consistency and magnitude of these findings underscore the importance of considering psychological, nutritional, and sleep-related factors as key determinants of maternal mental health, alongside more traditional clinical and demographic risk factors [19].

Furthermore, this thesis demonstrated the feasibility and utility of applying machine learning algorithms to complex, multi-dimensional datasets to identify predictive models of maternal mental health outcomes [32]. By leveraging techniques such as logistic regression, random forests, and gradient boosting, we were able to develop models with high accuracy,

sensitivity, and specificity for distinguishing between women with and without postpartum mental health problems [35]. These models can serve as a foundation for the development of screening tools and risk stratification approaches to identify women in need of additional support and intervention. However, it is important to acknowledge the limitations of this thesis, including the relatively small sample size for some analyses, the cross-sectional nature of the data, and the reliance on self-report measures for key exposures and outcomes [1]. These limitations highlight the need for further research with larger, longitudinal datasets, more objective measurement approaches, and more rigorous designs to establish causal relationships between maternal health factors and postpartum mental well-being [37].

In conclusion, this thesis provides compelling evidence for the role of psychological, nutritional, and sleep-related factors in shaping maternal mental health outcomes, and underscores the importance of a holistic, multi-disciplinary approach to promoting maternal well-being [29].

By integrating screening and intervention strategies that target these key domains of health, we can work towards a future in which all women have the support and resources they need to thrive during the critical perinatal period and beyond. Our findings also lay the groundwork for further research to refine our understanding of the complex determinants of maternal and child health outcomes and to develop more personalized, effective approaches to prevention and treatment. Ultimately, by investing in the well-being of mothers, we can promote healthier, more resilient families and communities for generations to come.

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Appendix

Table 6: Description of key variables used in the analysis

Variable	Type	Scale/Range	Description
MH_outcome	Binary	0 = No, 1 = Yes	Maternal mental health outcome
V4AH01	Categorical	1 = Breastfeed only, 2 = Bottle feed only, 3 = Both	Baby feeding type
Stress_average	Numerical	0-40	Average stress score from the Perceived Stress Scale
HassleUpliftIntensity	Numerical	0-1	Ratio of hassles intensity to uplifts intensity from the Pregnancy Experience Scale
ResilienceLevel	Ordinal	1 = Low (< 50), 2 = Moderate (50-75), 3 = High (> 75)	Resilience level based on the Connor-Davidson Resilience Scale score
AHEI2010	Numerical	0-110	Alternate Healthy Eating Index score
DT_PROT	Numerical	0-200	Protein intake (g)
DT_CARB	Numerical	0-500	Carbohydrate intake (g)
DT_FOLAC	Numerical	0-1000	Folic acid intake (mcg)
DT_IRON	Numerical	0-100	Iron intake (mg)
DT_CALC	Numerical	0-2000	Calcium intake (mg)
sleep_dur_avg_all_Mod	Numerical	0-1440	Average sleep duration (minutes) from actigraphy
sleep_SE_avg_all_Mod	Numerical	0-100	Average sleep efficiency (%) from actigraphy
Continued on next page			

Table 6 – continued from previous page

Variable	Type	Scale/Range	Description
sleep_Frag_avg_all_Mod	Numerical	0-100	Average sleep fragmentation (%) from actigraphy
CRace	Categorical	1 = Non-Hispanic White, 2 = Non-Hispanic Black, 3 = Hispanic, 4 = Asian, 5 = Other	Maternal race/ethnicity
V1AF02	Ordinal	1 = 8th grade or less, 2 = 9th-12th grade (no diploma), 3 = High school graduate or GED, 4 = Some college credit (no degree), 5 = Associate/technical degree, 6 = Bachelor's degree, 7 = Master's degree, 8 = Doctorate or professional degree	Maternal education level
Continued on next page			

Table 6 – continued from previous page

Variable	Type	Scale/Range	Description
V1AF14	Ordinal	1 = Under \$5,000, 2 = \$5,000-\$9,999, 3 = \$10,000-\$14,999, 4 = \$15,000-\$19,999, 5 = \$20,000-\$24,999, 6 = \$25,000-\$34,999, 7 = \$35,000-\$49,999, 8 = \$50,000-\$74,999, 9 = \$75,000-\$99,999, 10 = \$100,000-\$149,999, 11 = \$150,000-\$199,999, 12 = \$200,000 or more	Annual household income
Age_at_V1	Numerical	13-45	Maternal age (years) at first study visit
DT_ALCO	Numerical	0-100	Alcohol intake (g)
DT_CAFFN	Numerical	0-1000	Caffeine intake (mg)
VITD_MCG	Numerical	0-100	Vitamin D intake (mcg)
TOTAL_CHOLINE	Numerical	0-1000	Total choline intake (mg)
DT_SUG_T	Numerical	0-500	Total sugar intake (g)
DT_FIBE	Numerical	0-100	Dietary fiber intake (g)
DT_CHOL	Numerical	0-1000	Cholesterol intake (mg)
DT_SFAT	Numerical	0-100	Saturated fat intake (g)
DT_MFAT	Numerical	0-100	Monounsaturated fat intake (g)
DT_PFAT	Numerical	0-100	Polyunsaturated fat intake (g)
DT_VITC	Numerical	0-1000	Vitamin C intake (mg)
Continued on next page			

Table 6 – continued from previous page

Variable	Type	Scale/Range	Description
DT_THIA	Numerical	0-10	Thiamin (Vitamin B1) intake (mg)
DT_RIBO	Numerical	0-10	Riboflavin intake (mg)
DT_VITB6	Numerical	0-10	Vitamin B6 intake (mg)
DT_VB12	Numerical	0-20	Vitamin B12 intake (mcg)
has_healthcare	Binary	0 = No, 1 = Yes	Access to healthcare
StressLevel	Binary	0 = Low, 1 = High	Stress level
ChronHTN	Binary	0 = No, 1 = Yes	Chronic hypertension
V2AJ01a1	Categorical	1-32	Type of physical activity or exercise (Activity 1)
V2AJ01a2	Numerical	0-1440	Duration of physical activity or exercise (Activity 1) in minutes
PRENATALAMOUNT	Ordinal	1 = None, 2 = 1-3 times/month, 3 = 4-6 times/week, 4 = 1 time/day, 5 = 2 or more times/day	Prenatal supplement intake frequency

Table 6: Description of key variables used in the analysis

Appendix A: Detailed Statistical Analysis

A.1 Sleep Health Metrics

A.2 Psychological Metrics

Parameter	Normal	Adverse	P-value	Effect Size	95% CI
Sleep Duration (min)	512.4 $\pm$ 45.2	521.7 $\pm$ 54.8	0.2	0.2	[-5.1, 23.7]
Sleep Efficiency (%)	88.4 $\pm$ 4.1	87.1 $\pm$ 3.8	0.03*	-0.3	[-2.3, -0.1]
Sleep Fragmentation (%)	18.6 $\pm$ 5.7	20.7 $\pm$ 5.6	0.01*	0.4	[0.5, 3.6]
WASO (min)	52.3 $\pm$ 19.5	58.5 $\pm$ 20.8	0.03*	0.3	[0.5, 11.9]
* Indicates statistical significance at $p < 0.05$					

Table 7: Sleep Parameter Analysis Between Normal and Adverse Outcome Groups

Parameter	Normal	Adverse	P-value	Effect Size	95% CI
Stress Average	1.5 $\pm$ 0.3	1.7 $\pm$ 0.3	< 0.001*	0.9	[0.2, 0.4]
Hassle:Uplift Ratio	0.6 $\pm$ 0.2	0.7 $\pm$ 0.3	< 0.001*	0.8	[0.1, 0.2]
Resilience Level	1.2 $\pm$ 0.4	1.5 $\pm$ 0.5	< 0.001*	0.6	[0.1, 0.4]
* Indicates statistical significance at $p < 0.001$					

Table 8: Psychological Parameter Analysis Between Outcome Groups